$\Box A^\mu = 0$

$\partial_\mu A^\mu = 0$

$A^\mu = Re(a_0^\mu e^{ik_\nu x^\nu})$

$k^\nu = (\frac{\omega}{c}, \bar{k})$

$\tilde{a}_0^\nu = a_0^\nu + \alpha k^\nu$

$\tilde{a}_0^\nu k_\nu = a_0^\nu k_\nu + \alpha k^\nu k_\nu = 0$

$\longrightarrow$ $k_\nu k^\nu = 0$, $(\frac{\omega}{c})^2 - \bar{k}^2 = 0$

$k_\nu a_0^\nu = 0$

$e^{ik_\nu x^\nu} = e^{i\omega t - i\bar{k}\bar{r}}$

1 Переход к условию $\varphi = 0$, $div\bar{A} = 0$

ось $x \parallel \bar{k}$



$k^\nu = (\frac{\omega}{c}, \frac{\omega}{c}, 0, 0)$, $a_0^\nu k_\nu = 0 \Rightarrow \frac{\omega}{c}\varphi - \frac{\omega}{c}A_x = 0$

$A_x = \varphi$

пост. в-р $a_0^\nu = (\varphi, A_x, A_y, A_z) = (\varphi, \varphi, A_y, A_z)$

$\tilde{a}_0^\nu = a_0^\nu + \alpha k^\nu$, $\alpha = -\frac{\varphi c}{\omega}$

$\tilde{a}_0^\nu = (0, 0, \tilde{A}_y, \tilde{A}_z)$

$\exists$ СО, в которой $\varphi = 0$, $\bar{A} \perp \bar{k}$, $div\bar{A} = 0$ (не зав. Лор. инвар.)

$\bar{A}(t, \omega) = Re(\bar{a}_0 e^{-i\omega t + ik\bar{r}})$

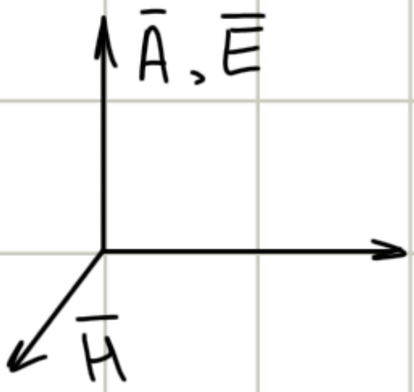
$\bar{E} = -\frac{1}{c}\frac{\partial \bar{A}}{\partial t} = Re(+\frac{i\omega}{c}\bar{a}_0 e^{-i\omega t + ik\bar{r}}) = Re(\bar{E}_0 e^{-i\omega t + ik\bar{r}})$, $\bar{E}_0 = \frac{i\omega}{c}\bar{a}_0$

$\bar{H} = rot\bar{A} = Re(i[\bar{k}, \bar{a}_0] e^{-i\omega t + i\bar{k}\bar{r}}) = Re(\bar{H}_0 e^{-i\omega t + i\bar{k}\bar{r}})$, $\bar{H}_0 = i[\bar{k}, \bar{a}_0]$

$\bar{a}_0 \perp \bar{k}$, $\bar{E}_0 = \frac{i\omega}{c}\bar{a}_0 \perp \bar{k}$, $\bar{H}_0 = i[\bar{n}_k \frac{\omega}{c}, \bar{a}_0] = [\bar{n}_k, \frac{i\omega}{c}\bar{a}_0] = [\bar{n}_k, \bar{E}_0]$

$\Downarrow$

$|\bar{H}_0| = |\bar{E}_0|$



$\{\bar{k}, \bar{E}, \bar{H}\}$ - правая тройка

поперечная волна

Фаза E: $-\omega t + k\bar{r} + \varphi$

$-\omega t + k\bar{r} + \varphi = 0 \longrightarrow (\bar{k}, \bar{r}) = \omega t - \varphi$



м-ть - пов-ть пост. фазы

плоская волна

2. Поляризация плоской поперек. ЭМ волны

$\bar{E} = Re(\bar{E}_0 e^{-i\omega t + i\bar{k}\bar{r}})$

$\bar{E}_0^2 = (\bar{E}_0, \bar{E}_0)$ - комплекс. число с фазой -2α

представим модуль $\bar{E}_0^2$ как $\bar{b}_0^2$, $\bar{b}_0$ - компл. в-р

$$\bar{E}_0^2 = \bar{b}_0^2 e^{-2i\alpha}$$

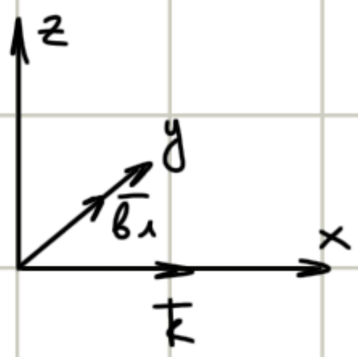
$$\bar{b}_0 = \bar{b}_1 + i\bar{b}_2$$

$$\bar{b}_0^2 = (\bar{b}_1 + i\bar{b}_2, \bar{b}_1 + i\bar{b}_2) = \bar{b}_1^2 - \bar{b}_2^2 + 2i(\bar{b}_1, \bar{b}_2)$$ - действ. пол. число

$$\Rightarrow (\bar{b}_1, \bar{b}_2) = 0, \quad \bar{b}_1^2 > \bar{b}_2^2$$

$$\bar{E}_0 = \bar{b}_0 e^{-i\alpha}$$

$$\bar{E} = Re\left((\bar{b}_1 + i\bar{b}_2) e^{-i\omega t + i\bar{k}\bar{r} - i\alpha}\right) = \bar{b}_1 \cos(\omega t - \bar{k}\bar{r} + \alpha) - \bar{b}_2 \sin(\omega t - \bar{k}\bar{r} + \alpha)$$



$\bar{k} \parallel \bar{x}$, $\bar{b}_1 \parallel \bar{y}$, $\bar{b}_2 \parallel \bar{z}$ или $\bar{b}_2 \parallel -\bar{z}$

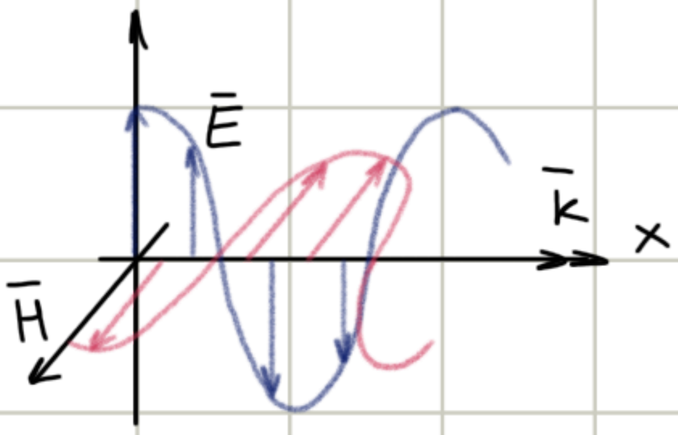
$$E_y = b_1 \cos(\omega t - \bar{k}\bar{r} + \alpha)$$

$$E_z = \pm b_2 \sin(\omega t - \bar{k}\bar{r} + \alpha)$$

$$\frac{E_y^2}{b_1^2} + \frac{E_z^2}{b_2^2} = 1$$ эллипт. поляриз.

$\bar{b}_1 = 0$ или $\bar{b}_2 = 0$ лин. поляриз.

fix t:



$b_1 = b_2 = b$, $E_y = b\cos(\dots)$, $E_z = \pm b \sin(\dots)$ круговая поляриз.

"+" - правая кр. пол.

"-" - левая кр. пол.

3. Связь w, $\bar{S}$ с комплексными амплитудами $\bar{E}_0$ и $\bar{H}_0$

$$w = \frac{\bar{E}^2 + \bar{H}^2}{8\pi}$$

$$\bar{E} = Re\left(\bar{E}_0 e^{-i\omega t + i\bar{k}\bar{r}}\right) = \frac{1}{2}\left(\bar{E}_0 e^{-i\omega t + i\bar{k}\bar{r}} + \bar{E}_0^* e^{i\omega t - i\bar{k}\bar{r}}\right)$$

$$\bar{E}^2 = \frac{1}{4}\left(\bar{E}_0 e^{-i\omega t + i\bar{k}\bar{r}} + \bar{E}_0^* e^{i\omega t - i\bar{k}\bar{r}}\right)\left(\bar{E}_0 e^{-i\omega t + i\bar{k}\bar{r}} + \bar{E}_0^* e^{i\omega t - i\bar{k}\bar{r}}\right) =$$

$$= \frac{1}{4}\left(2|\bar{E}_0|^2 + (\dots) e^{2i\omega t} + (\dots) e^{-2i\omega t}\right)$$

после уср. $T \gg \frac{1}{\omega}$ $\frac{|\bar{E}_0|^2}{2}$

$$\bar{H}^2 = [\text{после уср.}] = \frac{|\bar{H}_0|^2}{2} = \frac{|\bar{E}_0|^2}{2}$$

$$w = \frac{|\bar{E}_0|^2}{8\pi} = \frac{|\bar{H}_0|^2}{8\pi}$$

$$\bar{S} = \frac{c}{4\pi}[\bar{E},\bar{H}] = \frac{c}{4\pi}\frac{1}{4}[\bar{E}_0 e^- + \bar{E}_0^* e^+,\ \bar{H}_0 e^- + \bar{H}_0^* e^+] = \frac{c}{16\pi}\left([\bar{E}_0,\bar{H}_0^*] + [\bar{E}_0^*,\bar{H}_0]\right) =$$

$$= \frac{c}{16\pi}\left([\bar{E}_0,[\bar{n}_k,\bar{E}_0^*]] + [\bar{E}_0^*,[\bar{n}_k,\bar{E}_0]]\right) = \frac{c}{8\pi}|\bar{E}_0|^2\bar{n}_k = \frac{c}{8\pi}|\bar{H}_0|^2\bar{n}_k = wc\bar{n}_k$$

4. Ф-я Грина волн. ур-я

$$\Box A^\mu = \frac{4\pi}{c} j^\mu$$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)A^\mu = \frac{4\pi}{c} j^\mu$$

ф-я Грина волн. ур. $G(t-t', \bar{r}-\bar{r}')$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)G(t-t', \bar{r}-\bar{r}') = \delta(t-t')\delta(\bar{r}-\bar{r}')$$

Если G известна, то

$$A^\mu(t,\bar{r}) = \frac{4\pi}{c}\int dt' \int d^3\bar{r}'\ G(t-t', \bar{r}-\bar{r}')\, j^\mu(t',\bar{r}')$$

(+ решение в пустом пр-ве)

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)A^\mu = \frac{4\pi}{c}\int dt' \int d^3\bar{r}'\ \delta(t-t')\delta(\bar{r}-\bar{r}')\, j^\mu(t',\bar{r}') = \frac{4\pi}{c} j^\mu(t,\bar{r})$$

$$\frac{\partial}{\partial x^\mu}A^\mu = \frac{4\pi}{c}\int dt' \int d^3\bar{r}' \left[\frac{\partial}{\partial x^\mu} G(t-t', \bar{r}-\bar{r}')\right] j^\mu(t',\bar{r}') \ (=)$$

$$\| -\frac{\partial}{\partial x'^\mu} G(t-t', \bar{r}-\bar{r}')$$

$$(=) \ \text{- пов. инт.} \ (\to 0) + \frac{4\pi}{c}\int dt' \int d^3\bar{r}'\ G(t-t', \bar{r}-\bar{r}')\frac{\partial}{\partial x'^\mu} j^\mu(t',\bar{r}') = 0$$

$$G(t-t', \bar{r}-\bar{r}') = \frac{\delta\left(t' - \left(t - \frac{|\bar{r}-\bar{r}'|}{c}\right)\right)}{4\pi|\bar{r}-\bar{r}'|}$$

запазд. ф-я Грина

5. Запаздывающие потенциалы

$$A^\mu = \frac{1}{c}\int d^3\bar{r}'\ \frac{j^\mu\left(t - \frac{|\bar{r}-\bar{r}'|}{c}, \bar{r}'\right)}{|\bar{r}-\bar{r}'|}$$

$$\varphi(t,\bar{r}) = \int d^3\bar{r}'\ \frac{\rho\left(t - \frac{|\bar{r}-\bar{r}'|}{c}, \bar{r}'\right)}{|\bar{r}-\bar{r}'|}$$

$$A(t,\bar{r}) = \frac{1}{c}\int d^3\bar{r}'\ \frac{\bar{j}\left(t - \frac{|\bar{r}-\bar{r}'|}{c}, \bar{r}'\right)}{|\bar{r}-\bar{r}'|}$$